Where’s My Book?

SIFT-Based Object Detection in Cluttered Real-World Scenes

Project findings report

Classical Computer Vision Project • Team Work

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Dataset	Final metrics	Key takeaway
5 reference • 20 positive • 11 negative	Accuracy 90.32% • Precision 86.96% • Recall 100%	Multi-reference improved robustness; threshold tuning reduced false positives

1. Project Objective and Scope

Objective. The goal of this project was to detect a target red “Victoria Memorial” notebook in cluttered indoor scenes using only classical computer vision techniques. The detector had to distinguish the target book from distractor objects, including other notebooks with somewhat similar printed building structures.

- Input: reference views of the target notebook and scene images
- Output: target present / not present, with localization when confidently detected
- Constraint: to use feature-based matching rather than deep learning

2. Dataset Design

A small self-collected dataset was built to cover frontal views, perspective changes, scale changes, cluttered scenes, and hard negatives that visually resemble the target object.

Category	Count	Purpose
Reference	5	Clean target views used for matching
Positive	20	Scene images where the target book is present
Negative	11	Scene images where the target book is absent
Total	36	Complete working dataset

Figure 1. Five reference views used in the final multi-reference detector.

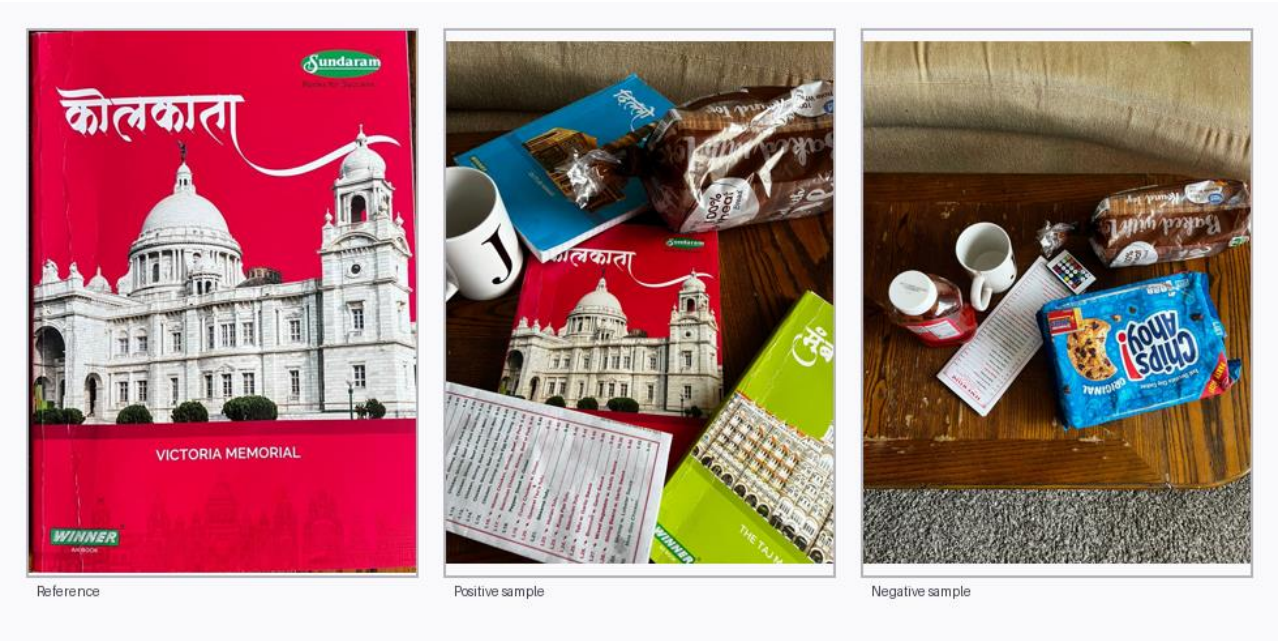

Figure 2. Representative dataset examples: a clean reference, a cluttered positive sample, and a negative sample.

3. Detection Pipeline

The final system followed a feature-based classical vision pipeline. Each test image was compared against all available reference images, and the best matching reference was selected using geometric consistency.

1. Converted images to grayscale.
2. Extracted SIFT key points and descriptors from both reference and test images.
3. Matched descriptors using brute-force matching and Lowe's ratio test.

4. Estimated homography with RANSAC when enough matches are available.
5. Counted homography inliers to measure geometric consistency.
6. Accepted a detection only when the match is strong enough and the inlier count exceeds a tuned threshold.

Final tuned decision rule:

Valid homography + at least 150 good matches + at least 120 inliers

4. Experimental Progression

Single-reference baseline. The initial version used only one reference image. It produced very strong results on the current dataset and showed that SIFT could reliably match the target notebook under clutter.

Multi-reference version. The system was then extended to use all five reference views. This increased robustness to viewpoint and scale changes, but it also introduced more false positives because some hard negatives contained notebooks with similar printed building structures.

Threshold tuning. To reduce these false positives, the final detector required both a minimum number of good matches and a minimum number of homography inliers. This improved precision without harming recall on the positive images.

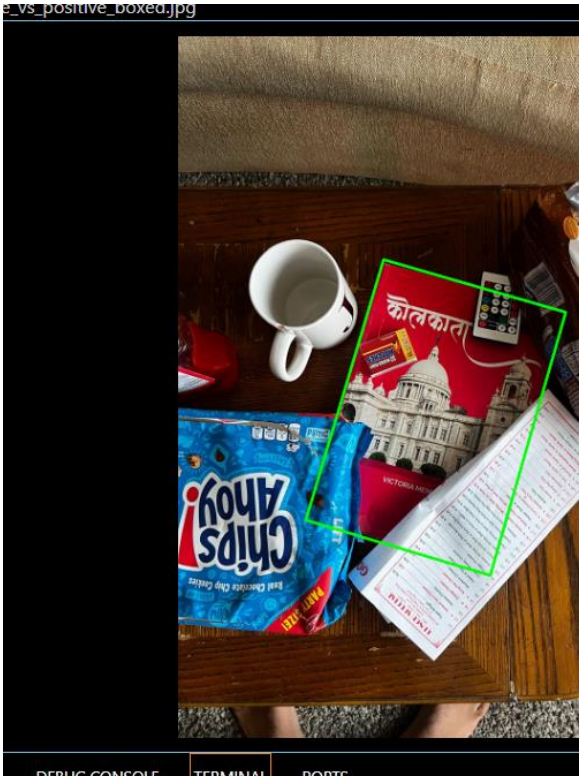

Figure 3. Example positive detection: the target notebook is localized with a green bounding box.

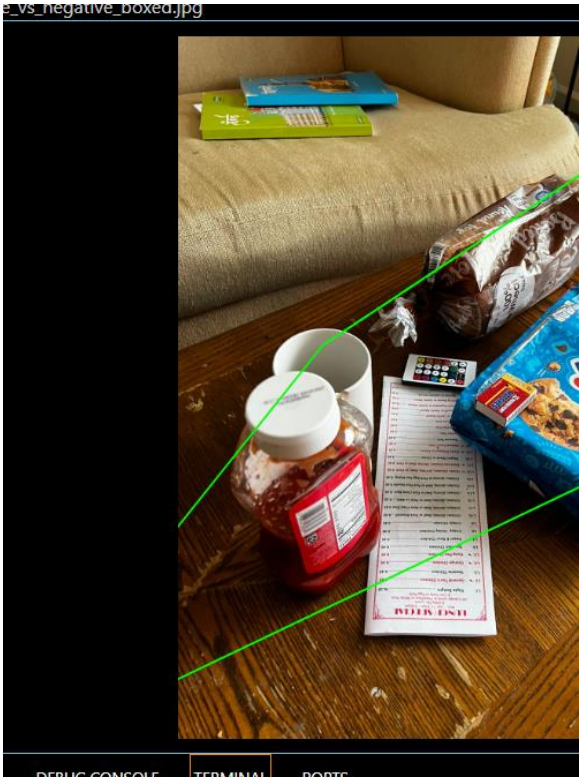

Figure 4. Example hard negative / false-positive behavior before stricter thresholding. Similar building patterns can still attract matches.

5. Final Quantitative Results

After multi-reference matching and threshold tuning, the detector achieved strong overall performance on the collected dataset.

True Positives (TP)	20
False Negatives (FN)	0
False Positives (FP)	3
True Negatives (TN)	8
Accuracy	0.9032
Precision / Recall	0.8696 / 1.0000

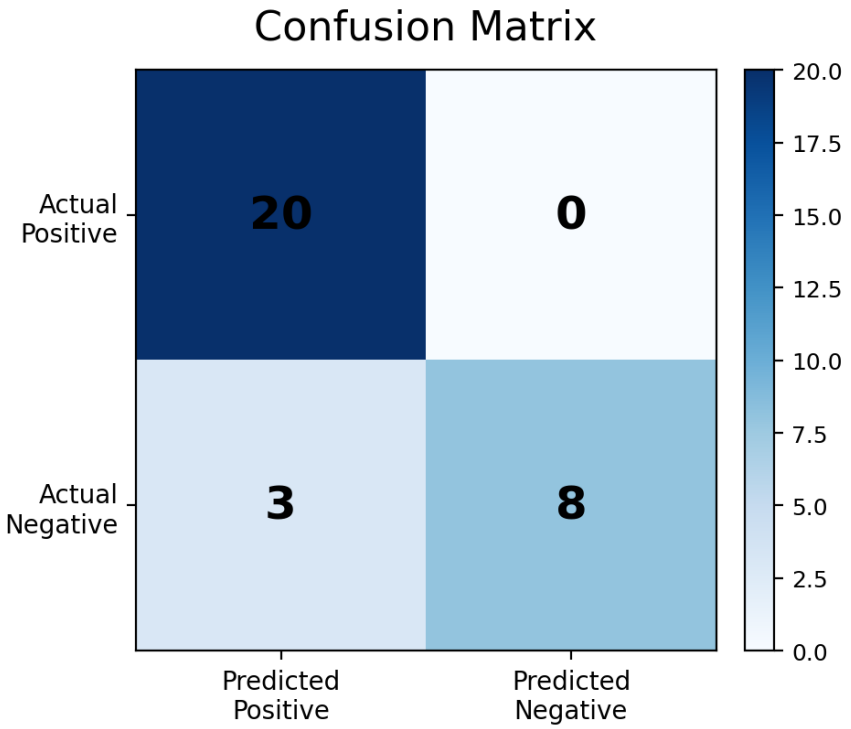

Figure 5. Final confusion matrix for the tuned multi-reference detector.

Interpretation. The detector found all 20 positive images (recall = 100%) and correctly rejected 8 of the 11 negative images. Only 3 negatives were falsely flagged as positives after tuning, which is a meaningful improvement over the untuned multi-reference version.

6. Findings and Limitations

- The strongest part of the system is recall: all target-containing images were detected in the final run.
- Using multiple references improved robustness because different scene images matched best with different reference views.
- Hard negatives remain challenging when they contain visually similar notebook covers with strong structural patterns.
- The current dataset is relatively small and captured in one indoor environment, so the reported performance should be viewed as proof-of-concept rather than guaranteed generalization.

7. Conclusion

This project successfully demonstrated that a SIFT-based classical vision pipeline can detect a specific target notebook in cluttered real-world scenes with strong performance. The work also showed an important practical lesson: adding more reference views improves flexibility, but threshold tuning is necessary to prevent false positives on visually similar distractors. The final system reached **90.32%** accuracy, **86.96%** precision, and **100%** recall on the collected dataset.

External Resources and Reflection

We used external resources such as Google Gemini, Stack Overflow, selected YouTube videos, and the course lecture slides to better understand SIFT, feature matching, homography, and the overall computer vision pipeline. These sources were used only for concept understanding and debugging ideas; the project code was written by us, and we did not use AI tools to generate the implementation. One of the most interesting parts of the project was working with SIFT and testing it on our own dataset rather than a ready-made one. We spent considerable time capturing images under different lighting conditions, angles, rotations, and cluttered scenes, and it was especially interesting to see how the detector behaved when three notebooks with similar building-style covers appeared in the scene. One of the more challenging parts was the homography-based detection step, where we had to verify geometric consistency before confirming a detection and drawing the bounding box. After some trial and error, along with help from Stack Overflow and video explanations, we understood the logic more clearly and completed the implementation ourselves.